

High-Performance Ultrapure Green CdSe/CdS Core/Crown Nanoplatelet Light-Emitting Diodes by Suppressing Nonradiative Energy Transfer

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Colloidal nanoplatelets (NPLs) are an emerging semiconductor nanocrystal in the display community due to their ultranarrow emission linewidth. Herein, an ultrapure green emitting nanocrystal light-emitting diode (LED) based on four-monolayer CdSe/CdS core/crown NPLs is developed. By applying the nonstacked nanoplates, the nonradiative energy transfer in the NPLs film is successfully suppressed. The nonstacked NPL-LEDs with pure green emission of 521.5 nm exhibit a low turn-on voltage of 2.1 V, a maximum luminance of 22 400 cd m⁻², a peak external quantum efficiency (EQE) of 2.16%, which is a sixfold enhancement comparing to the stacked NPL-LED (EQE = 0.34%). This work demonstrates the potential of core/crown NPLs for ultrawide color gamut displays.

1. Introduction

Following the remarkable progress in device performance and brightness, the external quantum efficiencies (EQEs) of quantum dots light-emitting diodes (LEDs) (QLEDs) have reached 30.9%, 25.04%, and 19.8% for red, green, and blue emission respectively at present, comparable to organic light emitting diodes.^[1–5] Meanwhile, an ultranarrow linewidth light source is demanded for future 8K high-definition display, augmented reality or true 3D holographic display. For instance, the primary color coordinates of Rec.2020 (International Telecommunication Union

-Radiocommunication Sector Recommendation BT.2020.) were defined by monochromatic laser light. Since human eyes are more sensitive to green color, a pure green light source becomes crucial to achieve a wide color gamut display. Due to the inhomogeneous broadening of quantum dot (QD) materials, the full width at half maximum (FWHM) of green QLED is about 30 nm. To the best of our knowledge, the narrowest ever FWHM of green QLED is 21 nm.^[6–8] Even for newly developed perovskite nanocrystals, the electroluminescence (EL) FWHM of green perovskite LED ranges from 18 to 23 nm.^[9–11] Hence, developing ultranarrow FWHM LEDs with pure green display is extremely essential for the next generation display.

Anisotropic cadmium chalcogenide nanoplatelets (NPLs) are a more promising material for achieving high color purity and wide color gamut. The atomically flat surface and extreme quantum confinement effect along the thickness of NPLs lead to high efficiency, an ultranarrow emission linewidth (≈ 10 nm), and a thickness-dependent emission (e.g., 460 nm for three-monolayer (ML) thickness, 515 nm for four-monolayer thickness, 550 nm for five-monolayer thickness).^[12–14] Such sharp emission is highly attractive in the high-level monochromatic LED industry for realizing vivid and high color saturation displays. Nowadays, various heterostructural NPLs based on CdSe core-only NPLs (e.g., core/shell, core/crown, and core/crown/shell NPLs) have been studied.^[12–14] For CdSe/CdS core/shell structure, CdS layers were grown on the top and the bottom of the CdSe core surface, forming a sandwich structure. But this inevitably causes an emission shifting and broadening since

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the electron will delocalize over the increased shell thickness. While for CdSe/CdS core/crown NPLs, the CdS layer was only extended in the lateral direction without any thickness and emission changes.^[15,16] Among the various heterostructural NPLs, four-monolayer CdSe/CdS core/crown NPL-LED is a good candidate for ultrapure green display because its color coordinates are right at the edge of green area in the chromatic diagram. To date, the efficiency of NPL-LED with red color is comparable to traditional QLED. The highest EQE of red-emitting NPL-LED was reported by Liu et al. in 2020. They exhibited a core/shell NPL-LED at 650 nm with FWHM = 26 nm and EQE = 19.32%.^[17] For green color NPL-LED, the highest EQE is 5% reported by Zhang et al. using five-monolayer (560 nm) CdSe/CdS core/crown nanoplatelets.^[18] However, it is inappropriate for pure green display considering their color coordinates (0.316, 0.651). For four-monolayer CdSe/CdS core/crown NPL-LED, unfortunately, the reported EQE is still no more than 0.5% and the luminance is only 4600 cd m⁻².^[19–21] Hence, it is of great significance to develop high efficiency four-monolayer CdSe/CdS core/crown NPL-LED for achieving ultrapure green light source and wide color gamut display.

In this work, we aim to improve the efficiency of four-monolayer CdSe/CdS core/crown NPL-LED by suppressing the nonradiative energy transfer caused by the stacked NPLs. By optimizing the antisolvent ratio in precipitation process, nonstacked NPLs were obtained. The as-prepared nonstacked NPLs devices exhibited a sixfold EQE enhancement comparing the stacked NPLs devices. The as-prepared NPL-LED possesses a narrow EL at 521.5 nm and an FWHM of 15 nm. A 2.16% peak EQE and a maximum luminance of 22 400 cd m⁻² have been reached. This work emphasized the potential applications of core/crown NPLs in the high-level monochromatic and

ultrawide color gamut displays. It also lays the foundation for further applying NPLs as emitting materials for LED with high efficiency.

2. Results and Discussion

The core/crown NPLs were prepared following a two-step synthesis method with some modification, which is provided in the Experimental Section. According to the photoluminescence (PL) and high-resolution transmission electron microscope image (Figure S1, Supporting Information), the four-monolayer CdSe core-only NPLs were synthesized successfully. Based on these core-only NPLs, the core/crown NPLs were subsequently prepared, of which the absorption and PL spectra are presented in Figure 1a. The two absorption peaks at 515 and 480 nm were the same as that in CdSe core-only NPLs, corresponding to light-hole exciton and heavy-hole exciton transition in CdSe core. The almost unchanged emission peak at 517.5 nm with an ultranarrow FWHM of 8.5 nm as well as the newly appeared absorption peak at 407 nm confirmed the growth of CdS crown layer along the lateral direction. The Stoke shift between the PL and first light-hole exciton absorption of core/crown NPLs was smaller than 3 nm, indicating a large emission and absorption overlap.

It is very common to find the self-stacking behavior of NPLs because of their large surface area and strong dipole–dipole interaction between two adjacent platelets.^[13,22–24] This stacking feature as well as the large emission and absorption overlap will cause serious energy transfer between NPLs. According to Burak's report, the homo-NRET (non-radiative energy transfer (NRET) between the same emitters) was serious in stacked NPLs,

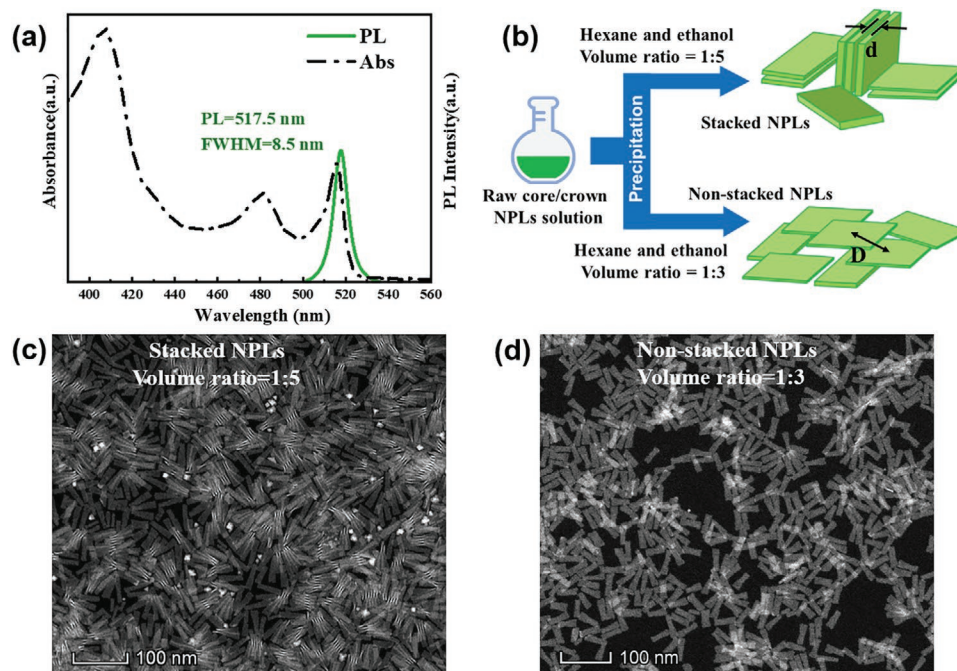


Figure 1. a) Absorption and photoluminescence spectra of core/crown NPLs. b) Schematic diagram of stacked NPLs and nonstacked NPLs prepared by different hexane and ethanol ratio (d and D refer to the distance between neighboring NPLs). c, d) TEM images of stacked and nonstacked NPLs.

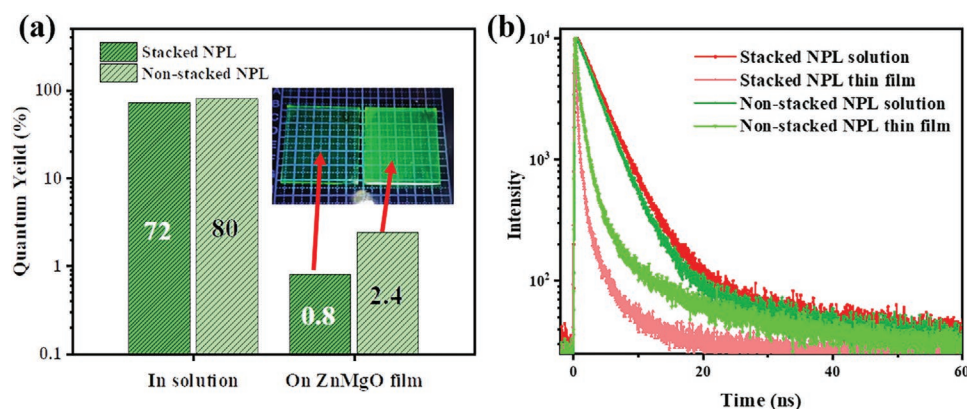


Figure 2. a) Quantum yield difference between stacked NPLs and nonstacked NPLs in solution and on ZnMgO film (inserted: photograph of stacked NPLs and non-stacked NPLs coated on ZnMgO film). b) PL decays of stacked NPLs and nonstacked NPLs in solution and on ZnMgO film.

which the NRET efficiency reached as high as 99.9%.^[22] To avoid the energy transfer and also eliminate the stacking problems, we redesigned the configuration of NPLs within film by optimizing the amount of ethanol as a polar solvent during NPLs precipitation (Figure 1b). It was reported that extra antisolvent (e.g., ethanol) will remove the ligands attached on the NPLs surface, forcing the NPLs to stack together to minimize their surface energy. We found that the critical ratio of hexane and ethanol was 1:5 for the first-time centrifugation to obtain stacked NPL, as shown in Figure 1c. For the ratio larger than 1:5, the less antisolvent caused weak attractions between NPLs (Figure S2, Supporting Information) and the suitable ratio ranged from 1:2 to 1:5. transmission electron microscope (TEM) images in Figure 1c,d confirmed that nonstacked NPLs were prepared successfully. By measuring the distance of Figure 1c,d, the average distances of stacked NPLs d and nonstacked NPLs D are 5.07 and 18.48 nm, respectively (Figure S3, Supporting Information). According to previous report, the Förster radius (distance of 50% energy transfer efficiency) of homo-NRET in 4ML NPLs was 13.48 nm.^[22] The distance between nonstacked NPLs (18.48 nm) was larger than 13.48 nm and was almost 3.5 times as long as that of stacked NPLs (5.07 nm), indicating a suppressed homo-NRET.

As shown in Figure 2a, the PL quantum yield (QY) of the stacked NPLs and nonstacked NPLs in solution were 72% and 80% respectively. But the QY dropped dramatically to 2.4% and 0.8% after spin-coated on ZnMgO film, which usually is adopted as an electron transfer layer in QLED devices. The QY of non-stacked NPLs film was two times higher than that of stacked NPLs, resulted from more severe nonradiative energy transfer in stacked NPLs. The photograph indicated a visual luminance difference of those films (Figure 2a). The atomic force microscope in Figure S4 in the Supporting Information indicated that the root mean square surface roughness of two films were almost the same ($RMS_{\text{stacked}} = 3.044$ nm while $RMS_{\text{nonstacked}} = 2.738$ nm), but the morphologies were quite different. Large-size clusters were noticed arising from the stacked NPLs while nonstacked NPLs film showed a flat and uniform surface. Moreover, the PL decay difference between stacked NPLs and nonstacked NPLs in solution and on ZnMgO film was also studied, shown in Figure 2b. The PL decays of stacked and nonstacked NPLs in solution are similar and the average exciton lifetimes are 4.36 and

4.03 ns, respectively. However, after spin-coated into thin film on ZnMgO, the stacked NPLs indicated a much faster PL decays lifetime (1.53 ns) comparing to the nonstacked NPLs (3.76 ns). It confirmed the nonradiative energy transfer in the stacked NPLs thin film was more serious than in nonstacked NPLs thin film, owing to that the NPLs were closely packaged together and the mean distance was shortened. Once exciton generated in the stacked NPLs, its energy will transfer between the closely stacked NPLs, and quickly be trapped by imperfect NPLs or the traps on other NPLs and quenched as nonradiative recombination.^[22,25] Hence, the nonstacked NPLs were expected to show better radiative recombination during the charge injection process.

A classical donor–acceptor energy transfer system was considered to explain the photoluminescent difference of NPL films and the corresponding PL decay. The energy transfer rate k_T between donor and acceptor (herein, both donor and acceptor are NPLs) can be described by the following equation^[26]

$$k_T = \frac{\kappa^2 \cdot \text{PLQY}}{\tau_D \cdot r^6} \cdot \frac{9 \ln 10}{128 \pi^2 N n^4} \cdot J \quad (1)$$

where κ^2 is the orientation factor describing the dipole–dipole interaction between NPLs, PLQY is the quantum yield of NPLs, τ_D is the lifetime of donor NPL, r is the distance between NPLs, N is the Avogadro's number, n is the refractive index of NPLs, and J is spectral overlap of emission and absorption.

The large spectral overlap of NPLs mentioned above leads to a high probability of the energy transfer between NPLs. In this case, only two factors will contribute to the energy transfer rate in NPLs since the donor and acceptor were the same materials. First of all, the value of κ^2 is related to the angle between two dipoles.^[26] For anisotropic NPLs, κ^2 is determined by the oriented arrangement of NPLs. $\kappa^2 = 5/4$ when NPLs were in a face-down state (comparatively more in nonstacked NPL films) and $\kappa^2 = 1/2$ when NPLs were in an edge-up state (comparatively more in stacked NPL films).^[27] Besides, distance r is a determining factor for energy transfer phenomenon since $k_T \propto 1/r^6$. In diluted solution, the distance of NPLs was long enough to avoid energy transfer. Once closely packaged into thin film, the distance between two stacked NPLs is much shorter than that of nonstacked NPLs. Hence, the short distance will lead to a strong

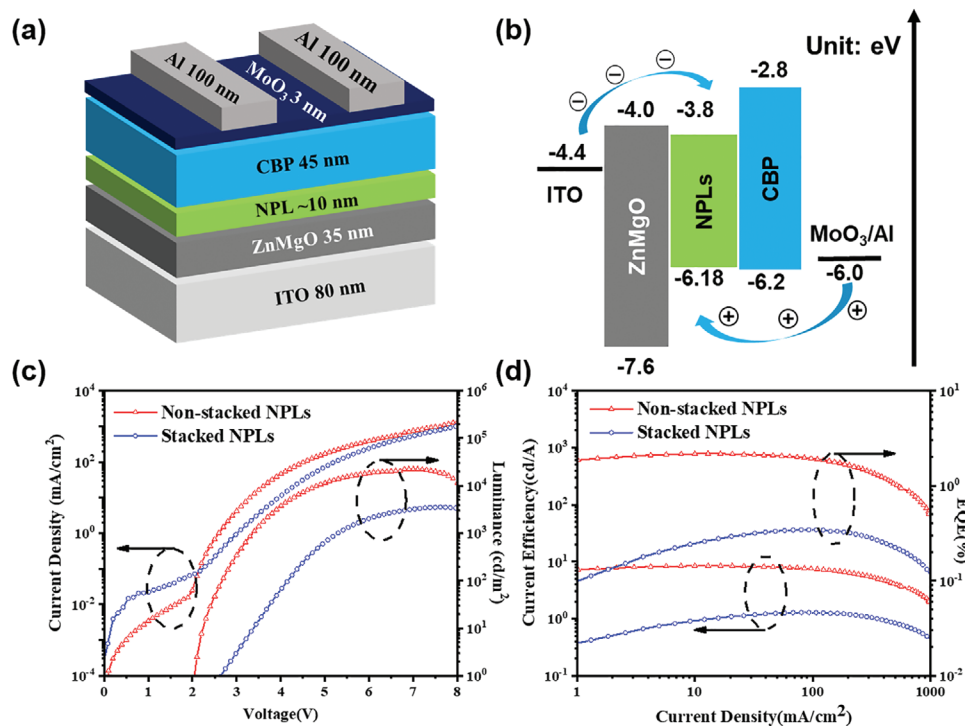


Figure 3. a) Device structure and b) energy level of devices. c) Current density–voltage–luminance (J – V – L) and d) current efficiency–current density–EQE (CE – J – EQE) curves of stacked and nonstacked NPLs devices.

nonradiative energy transfer between stacked NPLs. Substituting d and D obtained from the TEM images into r and using the photoluminescence quantum yield (PLQY) in solution, the maximum ratio of energy transfer rate between stacked NPLs and nonstacked NPLs is calculated to be $k_{TS}/k_{TN} \approx 1042$ (details shown in the Supporting Information). In reality, although the energy transfer ratio may not be so high since there is a fraction of face-down state NPLs in the stacked NPLs film, the edge-up state NPLs still dominate the nonradiative energy transfer. From the above, the high nonradiative energy transfer rate was the major cause of the poor PL performance of stacked NPLs film. When the stacked NPLs film was excited, the photogenerated exciton had a high probability to transfer around the neighboring stacked NPLs. Besides, since the stacking was caused by the reduction of surface ligand on NPLs, a portion of less passivated stacked NPLs would be the defect center and quenching the transferred exciton, leading to a low PLQY and a rapid nonradiative decay.^[25]

QLEDs were fabricated with a typical structure of ITO (indium tin oxide, 100 nm)/ZnMgO (Mg doped ZnO, 35 nm)/NPLs (10 nm)/CBP (4,4-*N,N'*-dicarbazole-1,1'-biphenyl, 45 nm)/MoO₃ (3 nm)/Al (100 nm). ZnMgO and NPL layers were first spin coated on the patterned ITO sequentially, then followed by thermal evaporation of CBP, MoO₃, and Al in a vacuum chamber. The device structure and energy diagram are shown in Figure 3a,b, respectively. According to the ultraviolet photon spectroscopy (UPS) measurement, the valance band maximum (VBM) of NPLs was calculated to be 6.18 eV (shown in Figure S5, Supporting Information). The VBM of CBP is comparable to that of NPL, while the conduction band minimum of CBP is 1 eV higher than that of NPL, providing efficient

hole transporting and electron blocking. ZnMgO was used as the electron transporting layer since it owned a high electron mobility ($\approx 10^{-3} \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$) and could efficiently block the extra holes.^[21] From the J – V – L curve in Figure 3c, the turn-on voltage (V_{on} , defined as the voltage when luminance equals to 1 cd m^{-2}) was reduced from 2.6 V for stacked NPL-LED to 2.1 V for nonstacked one. In addition to the low V_{on} , the nonstacked NPL devices exhibited significant enhancement of brightness and EQE, reaching $22\,400 \text{ cd m}^{-2}$ and $\text{EQE} = 2.16\%$, in comparison with 4739 cd m^{-2} and $\text{EQE} = 0.34\%$ for stacked NPL-LED. The NPL-LED device comparison has been listed in Table 1. Currently, the core/shell NPL-LED owns a highest device efficiency due to the fully passivated by outer shell. However, for core/crown NPL-LED, especially for pure green NPL-LED (520 nm), the efficiency is still far below other NPL-LED. The nonstacked NPLs have exhibited EQE at 1 order of magnitude higher than previous reports of pure green NPL-LED.

Comparing to the stacked NPL devices, nonstacked NPL devices showed a 4.7-fold and 6.3-fold improvement of brightness and EQE. Herein, we attributed the excellent device performance to the suppressed nonradiative energy transfer in nonstacked NPLs film according to the enhanced PLQY and longer PL decay. To verify this, the current–voltage (J – V) and capacitance–voltage (C – V) characterization are studied. Figure 4a presents the J – V characteristics of the two devices on double-logarithmic axes. The J – V curve was divided into several regions with a different power-law relationship ($J \propto V^n$), as reported similarly in refs. [31–33]. The index n indicates the states of the conduction (related to carriers) in device. When $n = 1$, the current is induced by the intrinsic traps or residual

Table 1. Comparison of different NPL-LED.

Material	Structure	EL [nm]	FWHM [nm]	EQE [%]	Maximum luminance [cd m ⁻²]
CdSe/Cd _{0.25} Zn _{0.75} S ^[17]	Core/shell	650	26	19.32	23 490
CdSe/CdS/CdZnS ^[28]	Core/shell/shell	634–648	21	9.92	46 000
CdSe/CdZnS ^[29]	Core/shell	658	–	8.39	1540
CdSe/Cd _{0.05} Zn _{0.95} S ^[30]	Core/shell	620	29	5.15	35 100
CdSe/CdZnS ^[31]	Core/shell	646	26	0.63	4499
CdSe/CdSeTe ^[14]	Core/crown	599	>40	3.57	34 520
CdSe/CdS ^[18]	Core/crown	560	14	5	33 000
CdSe ^[19]	Core-only	520	12.5	–	100
CdSe/CdS ^[20]	Core/crown	520	12	0.016	210
CdSe/CdS ^[21]	Core/crown	521	15	0.416	4684
This work	Core/crown	521.5	15	2.16	22 400

charges under a low voltage. When $n = 2$, the current is dominated by the drift current under the electric field, named trap space charge limited current (T-SCLC) regime.^[34] In Figure 4a, the stacked NPL-LED showed a higher current density and a lower index $n = 1.2$ in T-SCLC regime, deviating the square law. This was caused by the larger traps density in stacked NPL, which was verified by the shorten exciton lifetime in previous time-resolved photoluminescence (TRPL) results in Figure 2b. The index n will dramatically increase after the turn-on voltage. Generally, it results from the filled-traps in single carrier device. But for dual-carrier device, besides the trap-filling, the increase of index n is dominated by carrier's recombination. In this recombination and trap-filled limited current (RTFLC) regime, nonstacked NPL-LED showed a higher index ($n = 12.5$) than that of the stacked NPL-LED ($n = 8.5$), indicating a less trap density and a better charge recombination. Less trap density in nonstacked NPL-LED was in agreement with the T-SCLC regime and TRPL results, while better recombination matched the high luminance and EQE in Figure 3c,d. Furthermore, nonstacked NPL-LED exhibited a shorter-lasting RTFLC regime and obvious transition from the RTFLC to space-charge limited current (SCLC) regime, also indicating a less trap density in the nonstacked NPL-LED. In comparison, the reduction of surface

ligands in stacked NPL will lead to higher defects density and more traps in stacked NPL-LED. In other word, nonstacked NPL-LED owned a better charge transporting and recombination than stacked NPL-LED.

To further confirm the improvements of charge recombination, C–V measurement was also performed. The C–V curve of an emitting device can be defined by following equation^[35]

$$C(V) = \frac{dQ}{dV} = \frac{dQ_{in} + dQ_{trap} - dQ_r}{dV} \quad (2)$$

where Q_{in} , Q_{trap} , and Q_r are the charge quantity (induced by the perturbation AC signal V) of injected carriers, trapped carriers, and recombined carriers, respectively. Combining the equation and Figure 4b, the C–V curve can be divided into three regimes: neutral regime ($C = \text{constant}$), charge injection regime ($Q_r = 0, Q_{in} > 0$, C gradually increased to peak value), and recombination regime ($Q_r > 0$, C decreased rapidly). Compared with the C–V curve of stacked NPL-LED, that of nonstacked NPL-LEDs showed a smaller peak capacitance and peak-value voltage, indicating less traps in the emission layer. Furthermore, the slope of nonstacked NPL-LED in the recombination regime is steeper than that of the stacked NPL-LED, meaning a faster radiative

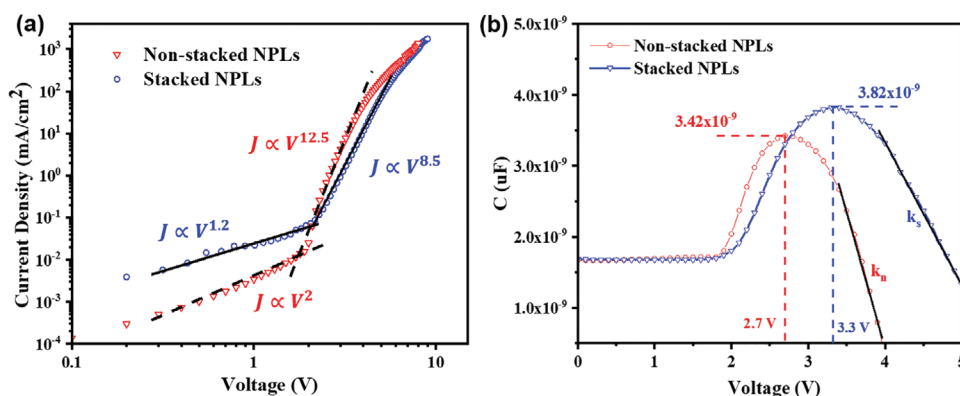


Figure 4. a) Current density–voltage characteristics (on double-logarithmic axes) and b) capacitance–voltage characteristics of stacked and nonstacked NPL-LEDs.

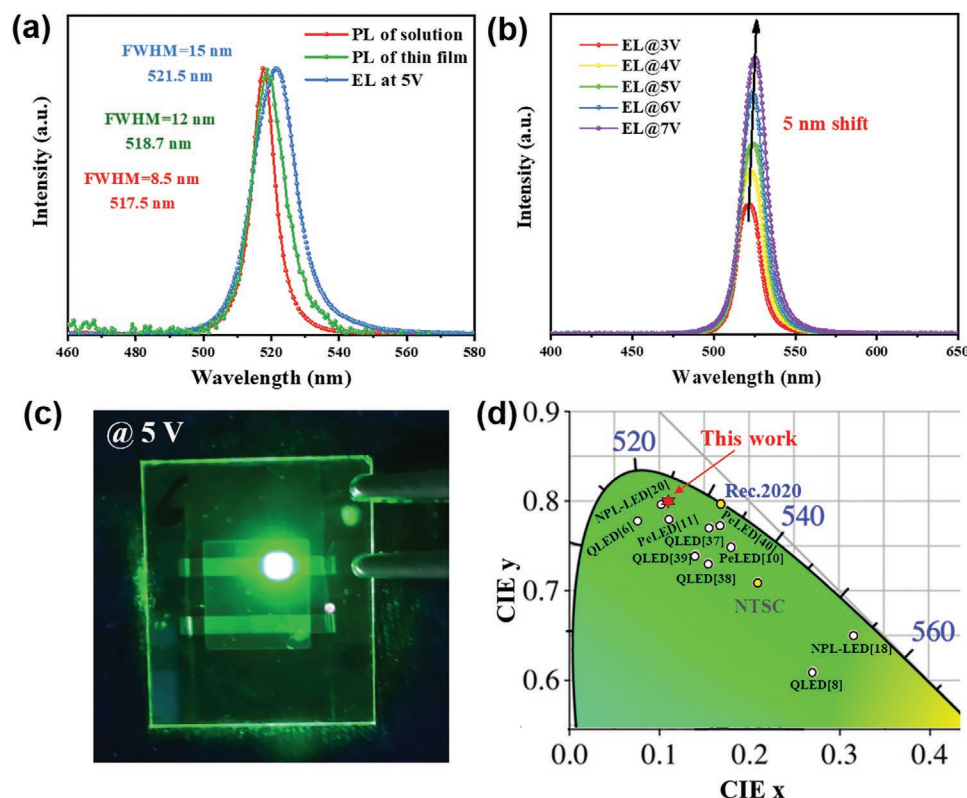


Figure 5. a) Spectrum comparison of NPLs solution, thin film, and EL device. b) EL spectrum of NPL-LED from 3 to 7 V. c) Photograph of the emitting NPL-LED. d) Color coordinates of reported green QLED, perovskite LED (PeLED), and NPL-LED.^[6,8,10,11,18,20,37–40]

charge recombination. Hence, for either photogenerated excitons or electrically injected carriers, nonstacked NPLs exhibited a better charge transporting and radiative recombination, leading to a high luminance and EQE.

Figure 5a shows the PL spectra of nonstacked NPLs in solution and in thin film and EL spectra of the corresponding NPL-LED. The slight broadening and redshifting of thin film resulted from the nonradiative energy transfer in closely packed nanocrystals. Both nonradiative energy transfer and electric field induced Stark effect contributed to the 4 nm redshift and 6.5 nm broadening.^[18,31,36] Figure 5b shows the EL spectra under different voltages. It can be seen that there is 5 nm peak wavelength shift while increasing the applied voltage from 3 to 7 V, confirming the electric field induced Stark effect. The photo of an operating NPL-LED at 5 V is presented in Figure 5c. The corresponding Commission Internationale de l'Eclairage coordinates were calculated to be (0.108, 0.800), shown in Figure 5d. For comparison purposes, the coordinates of previously reported green QLED, perovskite LED, and NPL-LED are also drawn (detailed data shown in Table S2, Supporting Information). Comparing to QLED and perovskite LED, the narrow linewidth and appropriate central wavelength enable the coordinates of NPL-LED be closer to the top of the chromatic diagram (520 nm). Combining the color coordinates of blue (0.131, 0.046) and red (0.708, 0.292) in Rec. 2020, the triangle color gamut area can reach 104% Rec. 2020, making NPL-LED attractive for ultrawide color gamut display.

3. Conclusions

In summary, high-efficiency pure green NPL-LED was fabricated by using nonstacked NPLs materials. The PL decay test indicated that the NRET in the nonstacked NPL was suppressed. The nonstacked NPL-LED exhibited a maximum luminance, a peak EQE and a high current efficiency up to 22 400 cd m⁻², 2.16%, and 8.7 cd A⁻¹, respectively. These results indicated a sixfold improvement of EQE comparing to the controlled group and 1 order of magnitude higher than previous reports of pure green NPL-LED. Furthermore, it implies that eliminating the self-stacking in anisotropic nanocrystals devices will reduce NRET and improve charge recombination, facilitating the device performance. The nonstacked NPL-LED demonstrated an emission peak at 521.5 nm with FWHM as narrow as 15 nm. The pure green spectra and high brightness signified the potential applications of core/crown NPLs in the high-level monochromatic and wide color gamut displays.

4. Experimental Section

Chemicals: Sodium myristate (≥99%), cadmium nitrate (99.0%), cadmium acetate dihydrate (≥98%), 1-octadecene (ODE, 90%), and oleic acid (OA, 90%), sulfur (S, 99.999%) were purchased from Sigma-Aldrich. Selenium powder (Se, ≥99.99%, 200 mesh) was purchased from Aladdin. Tri-*n*-octylphosphine (TOP, 97%) was purchased from Strem. CBP was purchased from Aglaia-tech. Methanol, ethanol, *n*-hexane, and toluene were purchased from Lingfeng.

Synthesis of Cadmium Myristate: 3130 mg of sodium myristate ($\geq 99\%$, Sigma-Aldrich) was added into 250 mL methanol with a strong stirring. 1230 mg of cadmium nitrate (99.0%, Sigma-Aldrich) was dissolved in 40 mL methanol. Then two solution were mixed and stirred to obtain white precipitate. After 1 h stirring, the precipitate was filtered and washed by methanol to wash away residual reactants. Finally, the precipitate was placed into vacuum drier at 60 °C for 12 h.

Synthesis of CdSe Core NPLs: 340 mg cadmium myristate and 20 mg Se were mixed with 30 mL ODE within a 100 mL flask. The mixture was degassed for three times at 80 °C. Then the mixture was heated to 240 °C under a continue argon flow. When temperature reached to 195 °C, 80 mg cadmium acetate dihydrate was swiftly dumped into the solution. The reaction was kept at 240 °C for 10 min. 1 mL OA was injected into the solution after the reaction. When cooled down to 100 °C, the solution was added into a mixed solution of hexane and ethanol (volume ratio = 1:3) and centrifuged at 5000 rpm for 10 min. The precipitate was dissolved in 14 mL hexane and recentrifuged at 5000 rpm for 5 min. The final supernatant was kept as the core nanoplatelets solution.

Preparation of CdS Precursor: 400 mg cadmium acetate dihydrate, 5 mL TOP and 5 mL ODE (90%, Sigma-Aldrich), 0.5 mL S-TOP (2.5 mol L⁻¹, sulfur dissolved in tri-*n*-octylphosphine stirring for 24 h at 100 °C) were mixed together and heated up to 80 °C under sustained stirring until forming a clear and transparent solution.

Synthesis of CdSe/CdS Core/Crown NPLs: 2.5 mL CdSe nanoplatelets core solution, 25 mL ODE, and 0.4 mL OA were mixed in a three-neck flask and degassed under vacuum. The temperature was set up to 240 °C under a continue argon flow. 5 mL CdS precursor solution was slowly injected during 195–220 °C. When reached to 240 °C, the reaction was kept for 5 min. For stacked NPLs, a mixed solution of hexane and ethanol (volume ratio > 1:5) was added into the solution at 100 °C and centrifuged at 8000 rpm for 8 min. The precipitation was dissolved in a mixed hexane and ethanol solution (volume ratio = 3:2) and centrifuged at 5000 rpm for 5 min. The precipitation was dissolved in octane and recentrifuged at 5000 rpm for 5 min. For nonstacked NPLs, a mixed solution of hexane and ethanol (volume ratio = 1:3) was added into the solution at 100 °C and centrifuged at 8000 rpm for 8 min. The precipitation was dissolved in a mixed hexane and ethanol solution (volume ratio = 3:2) and centrifuged at 5000 rpm for 5 min. The precipitation was dissolved in octane and recentrifuged at 5000 rpm for 5 min. The nanoplatelets solution was concentrated at 4 mg mL⁻¹.

NPL-LED Devices Fabrication: The NPL-LED was fabricated on ITO patterned glass substrates with a sheet resistance of 10 Ω sq⁻¹. The cleaning ITO substrate was cleaned by sonication in ITO cleaning agent, deionized water, and isopropanol for 15 min, respectively. ZnMgO (Mg doped ZnO, 20 mg mL⁻¹) was spin coated at 3000 rpm (1500 rpm min⁻¹ accelerated speed) for 45 s then annealing at 130 °C for 15 min. NPLs octane solution was deposited on ZnMgO at 1000 rpm (1500 rpm min⁻¹ accelerated speed) for 45 s. Then it was annealing at 100 °C for 5 min. Finally, CBP (45 nm), MoO₃ (3 nm), and Al (100 nm) were deposited by thermal evaporation at rate of 1, 0.1, 3 nm s⁻¹ under high vacuum ($< 5 \times 10^{-4}$ Pa). The luminance area of the NPL-LED is 2×2 mm².

Characterization of Material and Devices: The PL spectra and PLQY of NPLs film were measured by HAMAMATSU Absolute PL Quantum Yield Spectrometer C11347. Absorption spectra were recorded with PerkinElmer Lambda Spectrophotometer. TEM images were measured by TalosF200X. Scanning electron microscope images were measured by Zeiss Gemini 300. TRPL spectra were measured by FluoTime 300. UPS was measured by ESCALAB 250Xi.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords

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